INCHES POUND

The documentation process conversion measures necessary to comply with this revision shall be completed by 29 January 2000.

MIL-PRF-19500/159K
19 November 1999
SUPERSEDING
MIL-PRF-19500/159J
1 July 1998

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, VOLTAGE-REFERENCE,
TYPES 1N821-1, 1N823-1, 1N825-1, 1N827-1, AND 1N829-1, 1N821UR-1, 1N823UR-1,
1N825UR-1, 1N827UR-1, AND 1N829UR-1, JAN, JANTX, JANTXV, JANJ, JANS, JANHC AND JANKC;
RADIATION HARDENED (TOTAL DOSE ONLY) TYPES JANTXVM, D, L, R, F, G, H;
JANSM, D, L, R, F, G, H; JANHCM, D, L, R, F, G, H; AND JANKCM, D, L, R, F, G, H

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for 6.2 volts ± 5 percent, silicon, voltage-reference diodes. Five levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500, and two levels of product assurance are provided for each unencapsulated device type die. Seven levels of radiation hardened (total dose only) product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500, and two levels of product assurance for each unencapsulated device type die.

1.2 Physical dimensions. See figure 1 (similar to DO-7 and DO-35), figure 2 (similar to DO-213AA), figure 3 (JANHCA and JANKCA), figure 4 (JANHCB and JANKCB) and figure 5 (JANHCD and JANKCD).

1.3 Maximum ratings. (Unless otherwise specified, $T_A$ = +25°C).

| $P_T$ 1/ | $T_{STG}$ and $T_{op}$ | $I_{ZM}$ See note 1 | Power derating above $T_A$ = +25°C |
|---|---|---|---|
| mW | °C | mA dc | mW/°C |
| 500 | -55 to +175 | 70 | 3.33 |

NOTE;
1. To guarantee voltage temperature stability, it is necessary to maintain the proper $I_Z$ = 7.5 mA dc.

1.4 Primary electrical characteristics. Unless otherwise specified, primary electrical characteristics at $T_A$ = +25°C.

| Type | $\Delta V_Z$ (voltage temperature stability) | $Z_Z$ $I_Z$ = 7.5 mA dc | $V_Z$ $I_Z$ = 7.5 mA dc | | $I_R$ $V_R$ = 3.0 V |
|---|---|---|---|---|---|
| | | | Min | Max | |
| | mV dc | ohms | Volts | Volts | µA |
| 1N821-1, 1N821UR-1 | 96 | 15 | 5.89 | 6.51 | 2.0 |
| 1N823-1, 1N823UR-1 | 48 | 15 | 5.89 | 6.51 | 2.0 |
| 1N825-1, 1N825UR-1 | 19 | 15 | 5.89 | 6.51 | 2.0 |
| 1N827-1, 1N827UR-1 | 9 | 15 | 5.89 | 6.51 | 2.0 |
| 1N829-1, 1N829UR-1 | 5 | 15 | 5.89 | 6.51 | 2.0 |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service, 700 Robbins Avenue, Building 4D (DPM-DODSSP), Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Associated specification. The individual item performance requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1, 2, 3, 4, and 5 herein.

3.4.1 Lead finish. Unless otherwise specified herein, lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein.

3.5 Diode construction. These devices shall be constructed in a manner and using material which enable the diodes to meet the applicable requirements of MIL-PRF-19500 and this document.

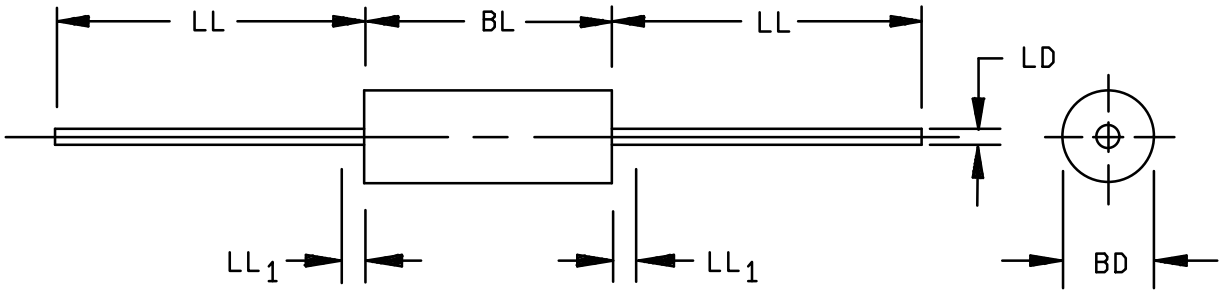
3.5.1 Dash-one construction. Shall be as specified in MIL-PRF-19500.

3.5.2 JANS construction. Construction shall be dash-one, category I or II metallurgical bond in accordance with MIL-PRF-19500.

3.5.3 JANHC and JANKC construction. JANHC and JANKC construction may differ in die size and bonding pad layout provided the manufacturing technology is identical (example: diffused junction, alloy junction).



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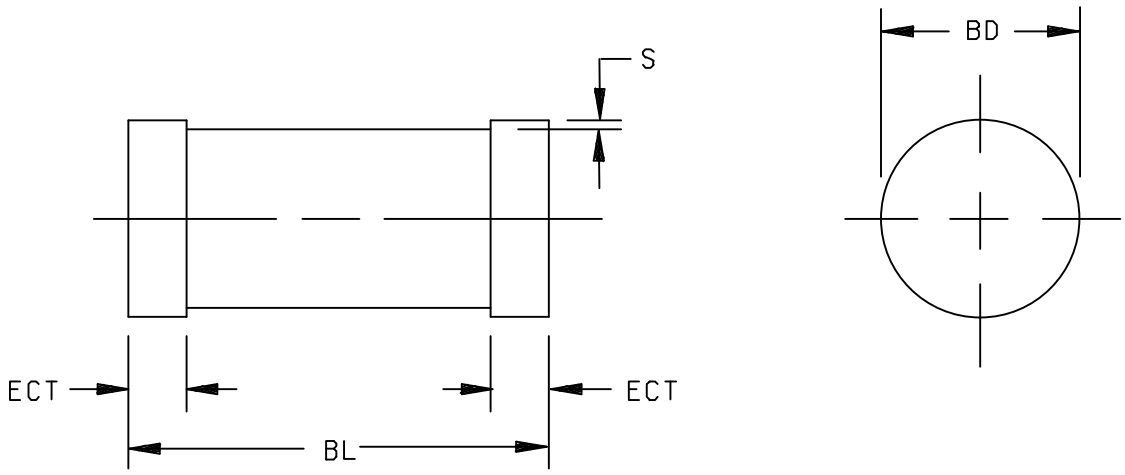

| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| BD | .060 | .107 | 1.52 | 2.72 | 3 |
| BL | .120 | .300 | 3.05 | 7.62 | 3 |
| LD | .018 | .023 | 0.46 | 0.58 | |
| LL | 1.000 | 1.500 | 25.40 | 38.10 | |
| $LL_1$ | --- | 0.050 | --- | 1.27 | 4 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Package contour optional within BD and length BL. Heat slugs, if any shall be included within this cylinder but shall not be subject to minimum limit of BD.
4. Within this zone, lead diameter may vary to allow for lead finishes and irregularities, other than heat slugs.

FIGURE 1. Physical dimensions, 1N821-1,through 1N829-1 (similar to DO-35 and DO-7).



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| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BD | .063 | .067 | 1.60 | 1.70 |
| BL | .130 | .146 | 3.30 | 3.71 |
| ECT | .016 | .022 | 0.41 | 0.56 |
| S | .001 Min | | 0.03 Min | |

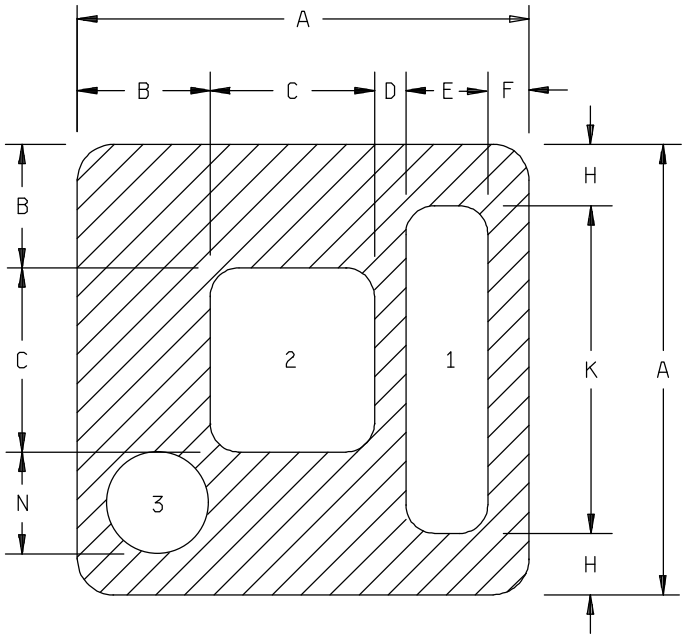
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Physical dimensions, 1N821UR-1, through 1N829UR-1 (DO-213AA).



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| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | .0280 | .0320 | 0.711 | 0.813 |
| B | .0080 | .0100 | 0.203 | 0.254 |
| C | .0104 | .0106 | 0.264 | 0.269 |
| D | .0019 | .0021 | 0.048 | 0.053 |
| E | .0054 | .0056 | 0.137 | 0.142 |
| F | .0020 | .0040 | 0.050 | 0.102 |
| G | .0280 | .0320 | 0.711 | 0.813 |
| H | .0030 | .0050 | 0.076 | 0.127 |
| J | .0030 | .0050 | 0.076 | 0.127 |
| K | .0209 | .0211 | 0.531 | 0.536 |
| L | .0080 | .0100 | 0.203 | 0.254 |
| M | .0104 | .0105 | 0.264 | 0.269 |
| N | .0059 | .0061 | 0.150 | 0.155 |




BACKSIDE MUST BE ELECTRICALLY ISOLATED TO ENSURE PROPER PERFORMANCE.

DESIGN DATA

Metallization:

Top: 1 (Cathode) . . . . Al
2 (Anode) . . . . . Al
3 (Test pad) . . . . Al

Circuit layout data:
For zener operation, cathode must be operated positive with respect anode.
Test pad is for wire bond evaluation only. No electrical contact is made with test pad.

Back .. Metalization. . . Au

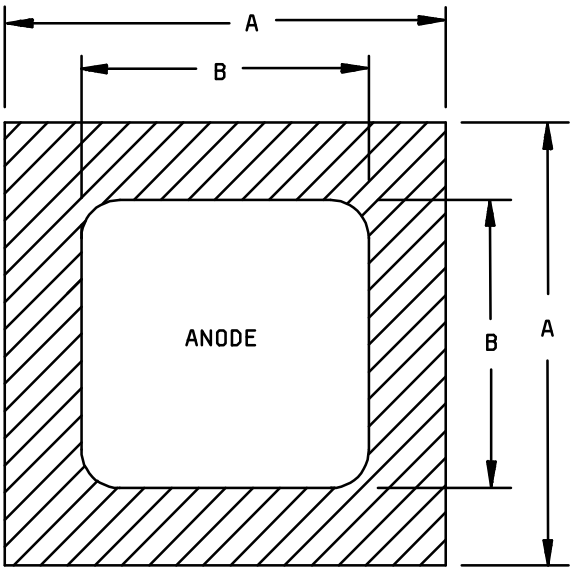
Al thickness . . . . . 25,000Å minimum
Gold thickness . . . . . 4,000Å minimum
Chip thickness . . . . . 0.010 inch (0.254 mm) ± 0.002 inch (0.051 mm).

NOTES:
1. Dimensions are in inches unless otherwise indicated.
2. Metric equivalents are given for general information only.

FIGURE 3. JANHC and JANKC (A-version) die dimensions.



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BACKSIDE IS CATHODE

| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | .024 | .037 | 0.61 | 0.94 |
| B | .014 | .029 | 0.36 | 0.74 |

DESIGN DATA

Circuit layout data:
For zener operation, cathode must be operated positive with respect anode. Back

Metallization:

Top: 2(Anode) . . . . . Al
1(Cathode) Al
. . . . . Au

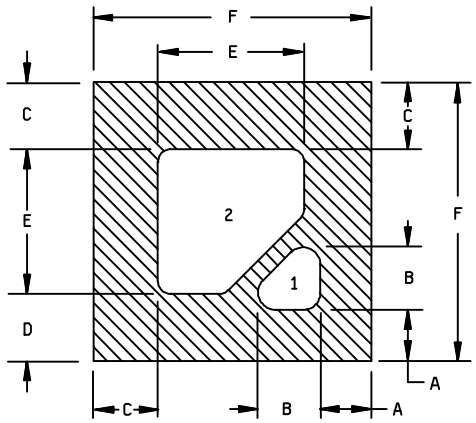
Al thickness . . . . . 40,000Å minimum
Gold thickness . . . . . 5,000Å minimum
Chip thickness . . . . . 0.010 inch (0.254 mm) ± 0.002 inch (0.051 mm).

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information.

FIGURE 4. JANHC and JANKC (B-version) die dimensions.



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BACKSIDE MUST BE ELECTRICALLY ISOLATED

| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | .0035 | .0065 | .088 | .165 |
| B | .0050 | .0080 | .127 | .200 |
| C | .0050 | .0065 | .127 | .165 |
| D | .0050 | .0065 | .127 | .165 |
| E | .0150 | .0165 | .381 | .419 |
| F | .0260 | .0290 | .660 | .737 |

DESIGN DATA

Metallization:

Circuit layout data: Top: (Anode) . . . Al
For zener operation, cathode must be operated positive with respect to anode. Back . . . . . Au

Al thickness . . . . . 40,000Å minimum
Gold thickness . . . . . 5,000Å minimum
Chip thickness . . . . . 0.010 inch (0.254 mm) ±0.002 inch (0.051 mm).

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information.
3. Backside must be electrically isolated.

FIGURE 5. JANHC and JANKC (C-version) die dimensions.



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3.6 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6.1 Marking of "UR" version devices. For "UR" version devices only, all marking (except polarity) may be omitted from the body, but shall be retained on the initial container.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4 and table I herein.

3.8 Electrical test requirements. The electrical test requirements shall be as specified in table I herein.

4 VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.1.1 Sampling inspection. Sampling inspection shall be in accordance with MIL-PRF-19500 and as specified herein, except that lot accumulation period shall be 3 months in lieu of 6 weeks.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-PRF-19500 and paragraph 4.4.5 herein.

4.2.2 JANHC and JANKC devices. Qualification for JANHC and JANKC devices shall be in accordance with MIL-PRF-19500.

4.2.3 JANJ devices. For JANJ level, 3.3.1 through 3.3.1.3 of MIL-PRF-19500 shall apply, except as modified herein. Supplier imposed requirements as well as alternate screens, procedures, and/or controls shall be documented in the QM plan and must be submitted to the Qualifying Activity for approval. When alternate screens procedures, and/or process controls are used in lieu of the JANJ screens herein equivalency shall be proven and documented in the QM Plan. Radiation characterization may be submitted in the QM plan at the option of the manufacturer, however, paragraph 3.3.1.1 of MIL-PRF-19500 is not required. Die lot controls and rework requirements shall be in accordance with MIL-PRF-19500 paragraphs 3.13 and D.3.13.2.1 for JANS level. Lot formation and conformance inspection requirements for JANJ shall be those used for JANTXV devices as a minimum.

4.2.4 JANJ Qualification. For JANJ qualification, 4.4.2.1 herein shall be performed as required by the Qualifying Activity.



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4.3 Screening (JANS, JANJ, JANTXV and JANTX levels only). Screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. Specified electrical measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see appendix E, table IV of MIL-PRF-19500) | JANS Level | JANJ Level | JANTXV and JANTX Level |
|---|---|---|---|
| 1a | Required | Not Required | Not Required |
| 1b | Required | Required | Required (JANTXV only) |
| 2 | Not Required | Not Required | Not Required |
| 3a | Required | Required | Required |
| 3b | Not Applicable | Not Applicable | Not Applicable |
| 3c | Not Applicable | Not Applicable | Not Applicable |
| 4 | Not Applicable | Not Applicable | Not Applicable |
| 5 | Not Applicable | Not Applicable | Not Applicable |
| 6 | Not Applicable | Not Applicable | Not Applicable |
| 7a | Not Applicable | Not Applicable | Not Applicable |
| 7b | Required | Required | Required |
| 8 | Required | Not Required | Not Required |
| 9 | Not Applicable | Not Applicable | Not Applicable |
| 10 | Not Applicable | Not Applicable | Not Applicable |
| 11 | Required<br>$V_Z$, $Z_Z$ | Required<br>$V_Z$, $Z_Z$ | Required<br>$V_Z$, $Z_Z$ |
| 12 | Required<br>See 4.3.1 | Required<br>t = 240 hours<br>See 4.3.1 | Required<br>See 4.3.1 |
| 13 | Required<br>Subgroups 2 and 3 of table I herein; $\Delta Z_Z \leq \pm 15$ percent of initial reading $T_A$ = +25°C ±2°C, $\Delta V_Z \leq \pm 0.004$ V dc from initial value for 1N821-1, 1N821UR-1, 1N823-1, 1N823UR-1, 1N825-1 1N825UR-1<br>$\Delta V_Z \leq \pm 0.003$ V dc from initial value for 1N827-1, 1N827UR-1, 1N829-1, 1N829UR-1 | Required<br>Subgroups 2 and 3 of table I herein; $\Delta Z_Z \leq \pm 15$ percent of initial reading $T_A$ = +25°C ±2°C, $\Delta V_Z \leq \pm 0.004$ V dc from initial value for 1N821-1, 1N821UR-1, 1N823-1, 1N823UR-1, 1N825-1 1N825UR-1<br>$\Delta V_Z \leq \pm 0.003$ V dc from initial value for 1N827-1, 1N827UR-1, 1N829-1, 1N829UR-1 | Required<br>Subgroups 2 of table I herein; $\Delta Z_Z \leq \pm 15$ percent of initial reading $T_A$ = +25°C ±2°C, $\Delta V_Z \leq \pm 0.004$ V dc from initial value for1N821-1, 1N821UR-1, 1N823-1,1N823UR-1, 1N825-1 1N825UR-1<br>$\Delta V_Z \leq \pm 0.003$ V dc from initial value for 1N827-1, 1N827UR-1, 1N829-1, 1N829UR-1 |
| 14a | Not Applicable | Not Applicable | Not Applicable |
| 14b | Optional 1/ | Optional 1/ | Optional 1/ |
| 15 | Required | Required<br>Attributes Data only<br>Film or Non-Film Techniques may be utilized | Not Required |
| 16 | Required | Required | Not Required |
| 17 | Not Required | Required<br>Subgroup 2 of table I herein | Not Required |

1/ Per MIL-PRF-19500



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4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

$I_Z$ = 7.5 mA dc, ±0.75 mA dc, $T_A$ = 150 +5°C, -0°C.

To better utilize burn-in equipment, higher values of $I_Z$ shall be permitted provided the junction temperature does not exceed +175°C.

4.3.2 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JAN, JANTX, JANTXV, and JANJ) of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the applicable steps of table I, subgroup 2 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500. For purposes of JANS inspection, a single device type shall be defined as devices from a single wafer lot (for each die type used in the construction). The conformance inspection sample shall be selected from the part category with the lowest $\Delta V_Z$ rating in the inspection lot.

| Subgroup | Method | Conditions |
|---|---|---|
| B1 | 2066 | |
| B2 | 2026<br>1022 | |
| B3 | 1056<br>4066<br>1071<br>2075 | Test Condition A, 25 Cycles<br>Not applicable<br>Test condition E. |
| B4 | 1037 | $I_Z$ = 35 mA dc at $T_A$ = +25°C; $t_{on}$ = $t_{off}$ = 30 seconds minimum for 4,000 cycles. Forced air cooling allowed during off cycle. |
| B5 | 1027 | $I_{ZM}$ = 70 mA dc for 96 hours. $T_A$ = +75°C or adjusted as required, to give an average lot $T_J$ = +200°C.±25°C |
| B6 | | Not applicable |

4.4.2.2 Group B inspection, appendix E, table VIb (JAN, JANTX, JANTXV and JANJ) of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B1 | 2026<br>1022 | |
| B2 | 1056<br>4066<br>1071 | Test Condition A, 25 Cycles<br>Not applicable<br>Test condition E |
| B3 | 1027 | (See 4.3.1) |
| B4 | 2075 | |
| B5 | | Not applicable |
| B6 | | Not applicable |



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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end-points) requirements shall be in accordance with the applicable steps of table I, subgroup 2 herein.

4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| C1 | 2066 | |
| C2 | 1056<br>2036<br>1071<br>1021 | Lead fatigue: Test condition E; tension: test condition A; 4 pounds weight, t = 15 ±3 seconds;<br>(lead fatigue and tension tests are not applicable to surface mount "UR" version devices).<br>Test Condition E<br>Omit initial conditioning |
| C3 | | Not applicable |
| C4 | 1041 | |
| C5 | | Not applicable |
| C6 | 1026 | $I_Z$ = 7.5 mA dc, $T_A$ = +100°C.(see 4.5.2) |
| C7 | | Not applicable |

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with table VIII of MIL-PRF-19500 and table II herein. MIL-PRF-19500, appendix F is not required when homogeneous product of radiation hard assembly (per applicable 36D form) is submitted and successfully passes group D sampling of total dose at an outside facility compliant to MIL-STD-750, method 1019. Submitted lots for group D sample inspection must be constructed using one homogeneous wafer lot for the zener and one wafer lot for the compensating die (die), as also described in the submitted 36D forms.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup tests in table IX of MIL-PRF-19500. Electrical measurements (end-points) requirements shall be in accordance with the applicable steps of table I, subgroup 2 herein.

4.4.5.1 Group E inspection, table IX of MIL-PRF-19500.

| Subgroup | Method | Conditions | Sample Plan |
|---|---|---|---|
| E1 | 1051 | 500 cycles | 22 devices, c = 0. |
| E2 | 1038 | Condition B, 1,000 hours (see 4.3.1) | 22 devices, c = 0. |
| E3 | | Not applicable | |
| E4 | | Not applicable | |
| E5 | | Not applicable | |
| E6 | | Not applicable | |



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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Voltage-temperature stability. The breakdown voltage of each diode type shall be measured and recorded at each of the specified temperatures. The lowest measured voltage shall be subtracted from the highest measured voltage for each diode. The difference value obtained shall not exceed the specified $\Delta V_Z$ per diode type.

4.5.2 Reference voltage time stability. The breakdown voltage shall be measured prior to life testing at 340 hours, and at the conclusion of the life test. The 340-hour reading shall be compared with the 0-hour reading and the 1,000-hour reading compared with the 340-hour reading. The change in breakdown voltage shall not exceed the limits specified. The test temperature for breakdown voltage shall be the same as the specified ambient life test temperature.

| Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Reference-voltage time stability | | $T_A$ = +100°C ± 2°C<br>(see 4.5.2 and 4.5.3)<br><br>$I_Z$ = 7.5 ± 0.01 mA dc | $\Delta V_Z$ | | | |
| 1N821-1, 1N821UR-1 | | (0 to 340 hours) | | | 7 | mV dc |
| 1N823-1, 1N823UR-1 | | | | | 7 | mV dc |
| 1N825-1, 1N825UR-1 | | | | | 7 | mV dc |
| 1N827-1, 1N827UR-1 | | | | | 6 | mV dc |
| 1N829-1, 1N829UR-1 | | | | | 5 | mV dc |
| 1N821-1, 1N821UR-1 | | (340 to 1,000 hours) | | | 4 | mV dc |
| 1N823-1, 1N823UR-1 | | | | | 4 | mV dc |
| 1N825-1, 1N825UR-1 | | | | | 4 | mV dc |
| 1N827-1, 1N827UR-1 | | | | | 3 | mV dc |
| 1N829-1, 1N829UR-1 | | | | | 3 | mV dc |

4.5.3 Reference voltage. The test current shall be applied until thermal equilibrium is attained (15 seconds minimum) prior to reading the reference voltage. For this test, the diode shall be suspended by its leads with mounting clips whose inside edge is located between 0.375 inch (9.53 mm) and 0.500 inch (12.70 mm) inch from the body and the mounting clips shall be maintained at the specified temperature. This measurement may be performed after a shorter time following application of the test current than that which provides thermal equilibrium if correlation to stabilized readings can be established to the satisfaction of the Government.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical Examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Reference voltage (see 4.5.3) | 4022 | $I_Z$ = 7.5 ± 0.01 mA dc | $V_Z$ | 5.89 | 6.51 | V dc |
| Small-signal breakdown impedance | 4051 | $I_Z$ = 7.5 ± 0.01 mA dc<br>$I_{sig}$ = 0.75 mA ac | $Z_Z$ | | 15 | Ω |
| Subgroup 3 | | | | | | |
| Voltage temperature stability (see 4.5.1 and 4.5.3) | | $I_Z$ = 7.5 ± 0.01 mA dc<br>$T_A$ = -55°C, 0°C. +25°C, +75°C, +100°C ± 2°C $I_Z$ = (see 1.4) | $\Delta V_Z$ | | | |
| 1N821-1, 1N821UR-1 | | | | | 96 | mV dc |
| 1N823-1, 1N823UR-1 | | | | | 48 | mV dc |
| 1N825-1, 1N825UR-1 | | | | | 19 | mV dc |
| 1N827-1, 1N827UR-1 | | | | | 9 | mV dc |
| 1N829-1, 1N829UR-1 | | | | | 5 | mV dc |
| Subgroups 4, 5, and 6 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 7 | | | | | | |
| Reverse current leakage | 4016 | DC method; $V_R$ = 3.0 V dc | $I_R$ | | 2.0 | µA |

1/ For sampling plan, see MIL-PRF-19500.



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TABLE II. Group D inspection.

| Inspection 1/ 2/ | MIL-STD-750 | | Symbol | Pre-irradiation limits | | Post-irradiation limits | | $\Delta V_{Z\theta}$ Pre-post $I_{RRAD}$ change | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|
| | Method | Conditions | | M, D, L, R, F, G, and H | | M, D, L, R, F, G, and H | | | | |
| | | | | Min | Max | Min | Max | Min | Max | |
| Subgroup 1 | | | | | | | | | | |
| Not applicable | | | | | | | | | | |
| Subgroup 2 | | $T_C$ = +25°C | | | | | | | | |
| Steady-state total dose irradiation | 1019 | $I_Z$ = 7.5 Condition A | | | | | | | | |
| Reference voltage (see 4.5.3) | 4022 | $I_Z$ = 7.5 ± 0.01 mA dc | $V_Z$ | 5.89 | 6.51 | 5.89 | 6.51 | | | V dc |
| Small-signal breakdown Impedance | 4051 | $I_Z$ = 7.5 $I_{stg}$ = 0.75 ac | $Z_Z$ | | 15 | | 15 | | | ohms |
| Reverse current leakage | 4016 | DC method; $V_R$ = 3.0 V dc | $I_R$ | | 2.0 | | 2.0 | | | µA |
| Voltage stability (see 4.5.3) | 4922 | $I_Z$ = 7.5 ± 0.01 mA dc $T_A$ = 25°C ± 2°C | $\Delta V_Z$ | | | | | | | |
| 1N821-1, 1N821UR-1 | | | | | | | | | ±3.0 | mV |
| 1N823-1, 1N823UR-1 | | | | | | | | | ±3.0 | mV |
| 1N825-1, 1N825UR-1 | | | | | | | | | ±2.0 | mV |
| 1N827-1, 1N827UR-1 | | | | | | | | | ±1.5 | mV |
| 1N829-1, 1N829UR-1 | | | | | | | | | ±1.0 | mV |

1/ For sampling plan, see MIL-PRF-19500.

2/ Group D qualification may be performed anytime prior to lot formation.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.4.1).

c. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCAM1N821) will be identified on the QPL.

| JANC ordering information | | | | | |
|---|---|---|---|---|---|
| PIN | Manufacturer | | | | |
| | 55801 | 55801 Radiation Designators M, D, L, R, F, G, H | 12954 | 12954 Radiation designators M, D, L, R, F, G, H | 12954 Radiation designators M, D, L, R, F, G, H |
| 1N821 | JANHCA1N821<br>JANKCA1N821 | JANHCA1N821<br>JANKCA1N821 | JANHCB1N821<br>JANKCB1N821 | JANHCB1N821<br>JANKCB1N821 | JANHCC1N821<br>JANKCC1N821 |
| 1N823 | JANHCA1N823<br>JANKCA1N823 | JANHCA1N823<br>JANKCA1N823 | JANHCB1N823<br>JANKCB1N823 | JANHCB1N823<br>JANKCB1N823 | JANHCC1N823<br>JANKCC1N823 |
| 1N825 | JANHCA1N825<br>JANKCA1N825 | JANHCA1N825<br>JANKCA1N825 | JANHCB1N825<br>JANKCB1N825 | JANHCB1N825<br>JANKCB1N825 | JANHCC1N825<br>JANKCC1N825 |
| 1N827 | JANHCA1N827<br>JANKCA1N827 | JANHCA1N827<br>JANKCA1N827 | JANHCB1N827<br>JANKCB1N827 | JANHCB1N827<br>JANKCB1N827 | JANHCC1N827<br>JANKCC1N827 |
| 1N829 | JANHCA1N829<br>JANKCA1N829 | JANHCA1N829<br>JANKCA1N829 | JANHCB1N829<br>JANKCB1N829 | JANHCB1N829<br>JANKCB1N829 | JANHCC1N829<br>JANKCC1N829 |



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6.5 Substitution of radiation hardened devices. See MIL-PRF-19500.

6.6 Substitution of $\Delta V_Z$ devices. Device types within this series with higher type numbers (lower $\Delta V_Z$) are a direct oneway substitution for lower type numbers (higher $\Delta V_Z$).

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5961-2096)



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**STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL**

**INSTRUCTIONS**

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| **I RECOMMEND A CHANGE:** | **1. DOCUMENT NUMBER**<br>**MIL-PRF-19500/159J** | **2. DOCUMENT DATE (YYMMDD)** |
|---|---|---|

**3. DOCUMENT TITLE** SEMICONDUCTOR DEVICE, DIODE, SILICON, VOLTAGE-REFERENCE, TYPES 1N821-1, 1N823-1, 1N825-1, 1N827-1, 1N829-1, 1N821UR-1, 1N823UR-1 1N825UR-1 1N827UR-1 1N829UR-1 JAN, JANTX, JANTXV, JANJ, JANS, JANHC AND JANKC; RADIATION HARDENED (TOTAL DOSE ONLY) TYPES JANTXVM, D, L, R, F, G, H; JANSM, D, L, R, F, G, H; JANHCM, D, L, R, F, G, H; JANKCM, D, L, R, F, G, H

**4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)**

**5. REASON FOR RECOMMENDATION**

| **6. SUBMITTER** | | |
|---|---|---|
| a. NAME (Last, First, Middle initial) | b. ORGANIZATION | |
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>Commercial<br>DSN<br>FAX<br>EMAIL | 7. DATE SUBMITTED<br>(YYMMDD) |

| **8. PREPARING ACTIVITY** | | | | |
|---|---|---|---|---|
| a. Point of contact: Alan Barone | b. TELEPHONE<br>Commercial | DSN | FAX | EMAIL |
| | 614-692-0510 | 850-0510 | 614-692-6939 | alan_barone@dscc.dla.mil |
| c. ADDRESS: Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad Street, Columbus, OH 43216-5000 | **IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:**<br>Defense Standardization Program Office (DLSC-LM)<br>8725 John J. Kingman Road, Suite 2533<br>Fort Belvoir, Vaginia 22060-6888<br>Telephone (703) 767-6888 DSN 427-6888 | | | |

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